

PERFORMANCE TASK 1 · LAB HANDOUT

Essential Linux Commands

IT 4112 — System Administration and Maintenance · Unit 0, Week 1–2

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Duration	3-hour lab + completion at home	Total points	100
Instructor	Mr. Ian Benitez	Deadline	

COMMAND LOGBOOK

#	Command run	What I expected	What actually happened
1	cd /	Expected to move to the root directory /.	The working directory changed to /.
2	ls -l	Expected to display the contents of / in long-listing format.	The root directory contents were displayed with permissions, owners, sizes, and dates.
3	cd etc	Expected to enter /etc using a relative path.	The working directory changed to /etc.
4	pwd	Expected to show the current directory.	It displayed /etc.
5	cd ../var/log	Expected to move from /etc to /var/log using a relative path.	The working directory changed to /var/log.
6	pwd	Expected to confirm that I was in /var/log.	It displayed /var/log.
7	ls -lt head	Expected to list files by newest modification time first.	The newest files in /var/log were shown first.
8	cd ~	Expected to return to my home directory.	The prompt returned to /home/nica.
9	cd -	Expected to return to the previous directory.	It returned to /var/log.
10	pwd	Expected to confirm the current directory.	It displayed /var/log.
11	whoami	Expected to display the current username.	It displayed nica.
12	id	Expected to display my UID, GID, and group memberships.	It displayed the account and group information for nica.
13	hostname	Expected to display the computer's hostname.	It displayed nikasvs.

14	<code>uname -a</code>	Expected to show Linux kernel and system information.	It showed Linux kernel information, including 5.15.0-187-generic and x86_64.
15	<code>cat /etc/os-release</code>	Expected to identify the Linux distribution and version.	It showed Ubuntu 22.04.5 LTS, Jammy Jellyfish.
16	<code>WHOAMI</code>	Expected to display my username.	It returned “WHOAMI: command not found” because Linux commands are case-sensitive.
17	<code>nproc</code>	Expected to show the number of available processing units.	It returned 2.
18	<code>lsblk</code>	Expected to show disks, partitions, sizes, and mount points.	It showed a 100G disk with the main partition mounted at /, plus loop devices and a ROM device.
19	<code>free -h</code>	Expected to show memory information in human-readable units.	It showed approximately 1.9 GiB total memory and 2.0 GiB swap.
20	<code>head -n 5 /proc/meminfo</code>	Expected to display the first five lines of memory information.	It showed MemTotal: 1986200 kB, along with MemFree, MemAvailable, Buffers, and Cached.
21	<code>cat /proc/uptime</code>	Expected to display the system uptime.	It returned 1957.62 seconds of uptime and the cumulative idle time.
22	<code>head -n 20 /proc/cpuinfo</code>	Expected to display processor information.	It showed GenuineIntel CPU information, model details, CPU speed, cache size, and CPU flags.
23	<code>head -n 5 /etc/passwd</code>	Expected to display the first five account records.	It displayed the records for root, daemon, bin, sys, and sync.
24	<code>wc -l /var/log/syslog</code>	Expected to count the number of lines in syslog.	It returned 1048.
25	<code>tail -n 20 /var/log/syslog</code>	Expected to display the last 20 lines of syslog.	It showed recent systemd, CRON, and apt-related activity.
26	<code>cat /etc/hostname</code>	Expected to display the configured hostname.	It displayed nika_svs.
27	<code>wc -l /etc/passwd</code>	Expected to count the lines in /etc/passwd.	It returned 33.
28	<code>mkdir -p isla-verde/{docs,logs,backups}</code>	Expected to create the isla-verde directory and its three subdirectories.	The directories were created.
29	<code>cd isla-verde</code>	Expected to enter the new project directory.	The working directory changed to ~/isla-verde.
30	<code>touch docs/inventory.txt</code>	Expected to create an empty inventory file.	inventory.txt was created inside docs.
31	<code>cp /etc/hosts backups/hosts.original</code>	Expected to copy /etc/hosts into the backups directory.	Nothing was printed, indicating silent success. The backup was later confirmed with <code>ls -lh</code> .

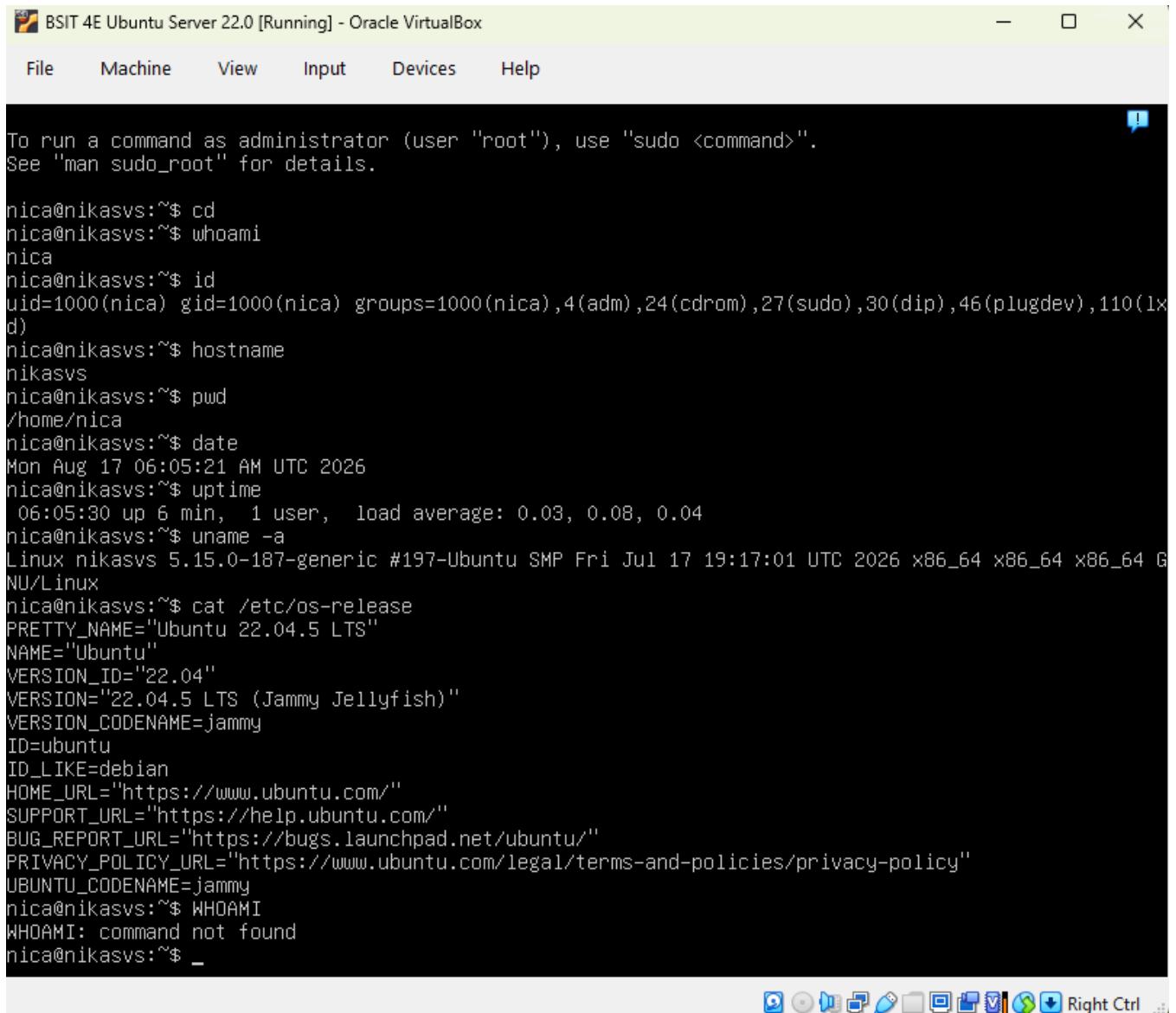
32	<code>cp /etc/os-release backups/os-release.o riginal</code>	Expected to create a backup of /etc/os-release.	Nothing was printed, indicating silent success. The backup was later confirmed with <code>ls -lh</code> .
33	<code>ls -lh backups/</code>	Expected to verify that the backup files existed and show their sizes.	It showed <code>hosts.original</code> and <code>os-release.original</code> .
34	<code>cd docs</code>	Expected to enter the docs directory.	The working directory changed to <code>~/isla-verde/docs</code> .
35	<code>mv inventory.txt inventory-2026.txt</code>	Expected to rename <code>inventory.txt</code> .	The file was renamed to <code>inventory-2026.txt</code> .
36	<code>cp -v inventory-2026.txt inventory.bak</code>	Expected to copy the inventory file and show the operation.	It displayed a message confirming the copy.
37	<code>rm -i inventory.bak</code>	Expected to ask for confirmation before deleting the backup.	It prompted for confirmation before deleting the file.
38	<code>ls -l</code>	Expected to show the remaining files in docs.	The directory listing showed the remaining inventory file.
39	<code>cut -d: -f7 /etc/passwd</code>	Expected to extract the login-shell field from each /etc/passwd entry.	It displayed shell paths such as <code>/bin/bash</code> , <code>/bin/false</code> , <code>/bin/sync</code> , and <code>/usr/sbin/nologin</code> .
40	<code>cut -d: -f7 /etc/passwd sort</code>	Expected to sort the extracted shells alphabetically.	The shell paths were grouped together alphabetically.
41	<code>cut -d: -f7 /etc/passwd sort uniq -c</code>	Expected to count how many accounts use each shell.	It showed 2 <code>/bin/bash</code> , 3 <code>/bin/false</code> , 1 <code>/bin/sync</code> , and 27 <code>/usr/sbin/nologin</code> .
42	<code>cut -d: -f7 /etc/passwd sort uniq -c sort -rn head -n 5</code>	Expected to rank the shell counts from highest to lowest and show the top five.	It showed <code>/usr/sbin/nologin</code> as the most common shell, followed by <code>/bin/false</code> , <code>/bin/bash</code> , and <code>/bin/sync</code> .
43	<code>find /etc -name "*.conf" wc -l</code>	Expected to count .conf files under /etc.	It returned 114.
44	<code>grep -n PermitRootLogin /etc/ssh/sshd_config</code>	Expected to search for <code>PermitRootLogin</code> and show line numbers.	It showed line 33: <code>#PermitRootLogin prohibit-password</code> and another commented reference at line 81.
45	<code>grep -v "^#" /etc/ssh/sshd_config grep -v "^\$"</code>	Expected to display active, non-comment, non-blank SSH configuration lines.	It displayed active settings such as <code>UsePAM yes</code> , <code>X11Forwarding yes</code> , and <code>PrintMotd no</code> .
46	<code>cd ~/isla-verde/docs</code>	Expected to return to the docs directory.	The prompt showed <code>~/isla-verde/docs</code> .
47	<code>date > recon.txt</code>	Expected to create <code>recon.txt</code> containing the current date and time.	The date and time were written to <code>recon.txt</code> .
48	<code>hostname >> recon.txt</code>	Expected to append the hostname without replacing the date.	<code>nikasvs</code> was appended to the file.
49	<code>uptime >> recon.txt</code>	Expected to append the uptime	The uptime and load average were

		information.	appended.
50	cat recon.txt	Expected to display all three pieces of reconnaissance information.	It displayed the date/time, hostname nikasvs, and uptime/load average.
51	man ls	Expected to open the manual page for ls.	The ls manual opened and showed its description and options.
52	apropos "disk space"	Expected to search the installed manuals for commands related to disk space.	It returned df (1): report file system disk space usage.
53	sudo cp /etc/hosts /etc/hosts.bak	Expected to create a backup before editing /etc/hosts.	The backup was created successfully.
54	sudo nano /etc/hosts	Expected to open /etc/hosts for editing with administrator privileges.	The entry 192.168.56.20 srv-isla01 was added.
55	diff /etc/hosts.bak /etc/hosts	Expected to show only the differences between the backup and edited file.	It showed 9a10,12 and the added host entry lines.
56	ping -c 2 srv-isla01	Expected to resolve the hostname and send two ICMP packets.	It resolved to 192.168.56.20. The VM showed timing and duplicate warnings and reported 50% packet loss, but a response was received.
57	sudo -l	Expected to show what commands my account can run using sudo.	It showed that nica may run (ALL: ALL) ALL.
58	sudo tail -n 20 /var/log/auth.log	Expected to show recent authentication and sudo activity.	It showed sudo commands run by nica, including the hosts backup, nano, and other commands.
59	cd ~/isla-verde	Expected to return to the isla-verde directory.	The prompt showed /isla-verde.
60	ls -l backups/	Expected to show the backup files and their permissions.	It showed hosts.original and os-release.original, initially with -rw-r--r-- permissions.
61	chmod 600 backups/hosts.original	Expected to restrict the backup so only the owner can read and write it.	The permissions changed to -rw-----.
62	ls -l backups/hosts.original	Expected to verify the new permissions.	It showed -rw----- 1 nica nica ... backups/hosts.original.
63	systemctl status ssh	Expected to check whether SSH is running and see its listening port.	SSH was active (running) and listening on port 22.

SCREENSHOTS EV-01 TO EV-06

EV-01: System Identity

Proves the VM username, hostname, kernel, and Ubuntu release were inspected



The screenshot shows a terminal window titled "BSIT 4E Ubuntu Server 22.0 [Running] - Oracle VirtualBox". The terminal displays the following commands and their outputs:

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

nica@nikasvs:~$ cd
nica@nikasvs:~$ whoami
nica
nica@nikasvs:~$ id
uid=1000(nica) gid=1000(nica) groups=1000(nica),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),110(lxd)
nica@nikasvs:~$ hostname
nikasvs
nica@nikasvs:~$ pwd
/home/nica
nica@nikasvs:~$ date
Mon Aug 17 06:05:21 AM UTC 2026
nica@nikasvs:~$ uptime
 06:05:30 up 6 min,  1 user,  load average: 0.03, 0.08, 0.04
nica@nikasvs:~$ uname -a
Linux nikasvs 5.15.0-187-generic #197-Ubuntu SMP Fri Jul 17 19:17:01 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
nica@nikasvs:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
nica@nikasvs:~$ WHOAMI
WHOAMI: command not found
nica@nikasvs:~$ _
```

This proves the VM username, hostname, Linux kernel version, and Ubuntu operating system version.

EV-02: Moving Around the Linux Filesystem

Proves absolute/relative navigation and use of pwd and ls.

```
nica@nikasvs:~$ cd /
nica@nikasvs:/$ ls -l
total 2097224
lrwxrwxrwx   1 root root          7 Sep 11  2024 bin -> usr/bin
drwxr-xr-x   3 root root        4096 Aug 10  06:46 boot
dr-xr-xr-x   2 root root        4096 Sep 11  2024 cdrom
drwxr-xr-x  19 root root        4020 Aug 17  05:59 dev
drwxr-xr-x  96 root root        4096 Aug 17  05:59 etc
drwxr-xr-x   3 root root        4096 Aug 17  05:59 home
lrwxrwxrwx   1 root root          7 Sep 11  2024 lib -> usr/lib
lrwxrwxrwx   1 root root         9 Sep 11  2024 lib32 -> usr/lib32
lrwxrwxrwx   1 root root         9 Sep 11  2024 lib64 -> usr/lib64
lrwxrwxrwx   1 root root        10 Sep 11  2024 libx32 -> usr/libx32
drwx-----  2 root root       16384 Aug 10  06:27 lost+found
drwxr-xr-x   2 root root        4096 Sep 11  2024 media
drwxr-xr-x   2 root root        4096 Sep 11  2024 mnt
drwxr-xr-x   2 root root        4096 Sep 11  2024 opt
dr-xr-xr-x 166 root root          0 Aug 17  05:59 proc
drwx-----  4 root root        4096 Aug 17  05:59 root
drwxr-xr-x  27 root root         780 Aug 17  06:03 run
lrwxrwxrwx   1 root root          8 Sep 11  2024/sbin -> usr/sbin
drwxr-xr-x   6 root root        4096 Sep 11  2024 snap
drwxr-xr-x   2 root root        4096 Sep 11  2024 srv
-rw-----  1 root root 2147483648 Aug 10  06:39 swap.img
dr-xr-xr-x  13 root root          0 Aug 17  05:59 sys
drwxrwxrwt  12 root root        4096 Aug 17  05:59 tmp
drwxr-xr-x  14 root root        4096 Sep 11  2024 usr
drwxr-xr-x  13 root root        4096 Sep 11  2024 var
nica@nikasvs:/$ _
```

```
nica@nikasvs:/$ cd etc
nica@nikasvs:/etc$ pwd
/etc
nica@nikasvs:/etc$ cd ../var/log
nica@nikasvs:/var/log$ pwd
/var/log
nica@nikasvs:/var/log$ ls -lt | head
total 1180
-rw-r----- 1 syslog adm          2777 Aug 17  06:17 auth.log
-rw-r----- 1 syslog adm       104542 Aug 17  06:17 syslog
-rw-rw-r--  1 root utmp       292292 Aug 17  06:03 lastlog
-rw-rw-r--  1 root utmp        1920 Aug 17  06:03 wtmp
drwxr-xr-x  2 landscape landscape  4096 Aug 17  06:03 landscape
-rw-rw----  1 root utmp         384 Aug 17  06:03 btmp
-rw-r----- 1 syslog adm       68634 Aug 17  05:59 cloud-init.log
-rw-r----- 1 root adm        4367 Aug 17  05:59 cloud-init-output.log
-rw-r--r--  1 root root       32032 Aug 17  05:59 faillog
nica@nikasvs:/var/log$ _
```

This proves that I successfully navigated the Linux directory tree using absolute and relative paths and verified my locations with pwd.

```
nica@nikasvs:/var/log$ cd ~
nica@nikasvs:~$ cd -
/var/log
nica@nikasvs:/var/log$ pwd
/var/log
nica@nikasvs:/var/log$ _
```

EV-03: Reading Files and System Information

Proves use of /etc/passwd, /var/log/syslog, and related read-only commands.

```
nica@nikasvs:/var/log$ cat /etc/hostname
nika_svs
nica@nikasvs:/var/log$ wc -l /etc/passwd
33 /etc/passwd
nica@nikasvs:/var/log$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
nica@nikasvs:/var/log$ _
```

```
nica@nikasvs:/var/log$ wc -l /var/log/syslog
1048 /var/log/syslog
nica@nikasvs:/var/log$ tail -n 20 /var/log/syslog
Aug 17 06:03:42 nikasvs systemd[1645]: Listening on REST API socket for snapd user session agent.
Aug 17 06:03:42 nikasvs systemd[1645]: Listening on D-Bus User Message Bus Socket.
Aug 17 06:03:42 nikasvs systemd[1645]: Reached target Sockets.
Aug 17 06:03:42 nikasvs systemd[1645]: Reached target Basic System.
Aug 17 06:03:42 nikasvs systemd[1]: Started User Manager for UID 1000.
Aug 17 06:03:42 nikasvs systemd[1]: Started Session 1 of User nica.
Aug 17 06:03:42 nikasvs systemd[1645]: Reached target Main User Target.
Aug 17 06:03:42 nikasvs systemd[1645]: Startup finished in 237ms.
Aug 17 06:04:24 nikasvs systemd[1]: Starting Download data for packages that failed at package install time...
Aug 17 06:04:24 nikasvs systemd[1]: update-notifier-download.service: Deactivated successfully.
Aug 17 06:04:24 nikasvs systemd[1]: Finished Download data for packages that failed at package install time.
Aug 17 06:14:24 nikasvs systemd[1]: Starting Cleanup of Temporary Directories...
Aug 17 06:14:24 nikasvs systemd[1]: systemd-tmpfiles-clean.service: Deactivated successfully.
Aug 17 06:14:24 nikasvs systemd[1]: Finished Cleanup of Temporary Directories.
Aug 17 06:17:01 nikasvs CRON[1700]: (root) CMD ( cd / && run-parts --report /etc/cron.hourly)
Aug 17 06:25:01 nikasvs CRON[1725]: (root) CMD (test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily ))
Aug 17 06:26:24 nikasvs systemd[1]: Starting Daily apt upgrade and clean activities...
Aug 17 06:26:27 nikasvs systemd[1]: apt-daily-upgrade.service: Deactivated successfully.
Aug 17 06:26:27 nikasvs systemd[1]: Finished Daily apt upgrade and clean activities.
Aug 17 06:26:27 nikasvs systemd[1]: apt-daily-upgrade.service: Consumed 2.843s CPU time.
nica@nikasvs:/var/log$
```

This proves that I used Linux commands to read configuration files, logs, account information, and system information without modifying them.

```
nica@nikasvs:/var/log$ cat /proc/uptime
1957.62 3818.26
nica@nikasvs:/var/log$ head -n 20 /proc/cpuinfo
processor       : 0
vendor_id      : GenuineIntel
cpu family     : 6
model          : 165
model name     : Intel(R) Core(TM) i5-10400 CPU @ 2.90GHz
stepping       : 3
microcode      : 0xe0
cpu MHz        : 2904.002
cache size     : 12288 KB
physical id    : 0
siblings       : 2
core id        : 0
cpu cores      : 2
apicid         : 0
initial apicid : 0
fpu            : yes
fpu_exception  : yes
cpuid level    : 22
wp             : yes
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mm
x fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_k
nwn_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch invpcid_single fsgsbase bmi1 avx2 bmi2 invpcid rdseed adx c
lflushopt arat md_clear flush_l1d arch_capabilities
nica@nikasvs:/var/log$ _
```

```
nica@nikasvs:/var/log$ head -n 5 /proc/meminfo
MemTotal:      1986200 kB
MemFree:       1469376 kB
MemAvailable:  1650696 kB
Buffers:       21640 kB
Cached:        286316 kB
nica@nikasvs:/var/log$
```

```
nica@nikasvs:/var/log$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	175Mi	1.4Gi	1.0Mi	329Mi	1.6Gi
Swap:	2.0Gi	0B	2.0Gi			

```
nica@nikasvs:/var/log$ nproc
2
nica@nikasvs:/var/log$ lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
loop0	7:0	0	63.9M	1	loop	/snap/core20/2318
loop1	7:1	0	87M	1	loop	/snap/lxd/29351
loop2	7:2	0	38.8M	1	loop	/snap/snapd/21759
sda	8:0	0	100G	0	disk	
└─sda1	8:1	0	1M	0	part	
└─sda2	8:2	0	100G	0	part	/
sr0	11:0	1	1024M	1	rom	

This proves the VM's CPU, memory, storage, and uptime information using commands such as nproc, free -h, lsblk, and /proc.

EV-04: Creating, Copying, and Deleting Safely

Proves CPU, memory, storage, and uptime were read from normal tools and /proc.

```
nica@nikasvs:/var/log$ cd ~
nica@nikasvs:~$ mkdir -p isla-verde/{docs,logs,backups}
nica@nikasvs:~$ cd isla-verde
nica@nikasvs:~/isla-verde$ touch docs/inventory.txt
nica@nikasvs:~/isla-verde$ _
```

```
nica@nikasvs:~/isla-verde$ cp /etc/hosts backups/hosts.original
nica@nikasvs:~/isla-verde$ cp /etc/os-release backups/os-release.original
nica@nikasvs:~/isla-verde$ ls -lh backups/
total 8.0K
-rw-r--r-- 1 nica nica 223 Aug 17 06:40 hosts.original
-rw-r--r-- 1 nica nica 386 Aug 17 06:40 os-release.original
```

```
nica@nikasvs:~/isla-verde$ cd docs
nica@nikasvs:~/isla-verde/docs$ mv inventory.txt inventory-2026.txt
nica@nikasvs:~/isla-verde/docs$ cp -v inventory-2026.txt inventory.bak
'inventory-2026.txt' -> 'inventory.bak'
```

```
nica@nikasvs:~/isla-verde/docs$ mv inventory.txt inventory-2026.txt
nica@nikasvs:~/isla-verde/docs$ cp -v inventory-2026.txt inventory.bak
'inventory-2026.txt' -> 'inventory.bak'
nica@nikasvs:~/isla-verde/docs$ rm -i inventory.bak
rm: remove regular empty file 'inventory.bak'? ls -l
```

This proves that I created the Isla Verde directories, copied files for backup, renamed a file, and used `rm -i` to delete safely.

EV-05: Finding Things, and Joining Tools Together

Proves the isla-verde structure and backup copies were created and verified.

```
nica@nikasvs:~/isla-verde/docs$ cd
nica@nikasvs:~$ cut -d: -f7 /etc/passwd
/bin/bash
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/bin/sync
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/bin/false
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/bin/false
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/bin/bash
/bin/false
nica@nikasvs:~$
```

```
nica@nikasvs:~$ cut -d: -f7 /etc/passwd | sort
/bin/bash
/bin/bash
/bin/false
/bin/false
/bin/false
/bin/sync
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
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/usr/sbin/nologin
/usr/sbin/nologin
/usr/sbin/nologin
nica@nikasvs:~$
```

```
nica@nikasvs:~$ cut -d: -f7 /etc/passwd | sort | uniq -c
  2 /bin/bash
  3 /bin/false
  1 /bin/sync
 27 /usr/sbin/nologin
nica@nikasvs:~$ _
```

```
nica@nikasvs:~$ cut -d: -f7 /etc/passwd | sort | uniq -c | sort -rn | head -n 5
 27 /usr/sbin/nologin
  3 /bin/false
  2 /bin/bash
  1 /bin/sync
```

```
nica@nikasvs:~$ sudo find /etc -name "*.conf" | wc -l
114
nica@nikasvs:~$ grep -n PermitRootLogin /etc/ssh/sshd_config
33:#PermitRootLogin prohibit-password
81:# the setting of "PermitRootLogin without-password".
nica@nikasvs:~$
```

```
nica@nikasvs:~$ grep -v "^#" /etc/ssh/sshd_config | grep -v "^$"
Include /etc/ssh/sshd_config.d/*.conf
KbdInteractiveAuthentication no
UsePAM yes
X11Forwarding yes
PrintMotd no
AcceptEnv LANG LC_*
Subsystem      sftp      /usr/lib/openssh/sftp-server
```

```
nica@nikasvs:~$ cd ~/isla-verde/docs
nica@nikasvs:~/isla-verde/docs$
nica@nikasvs:~/isla-verde/docs$ date > recon.txt
nica@nikasvs:~/isla-verde/docs$ hostname >> recon.txt
nica@nikasvs:~/isla-verde/docs$ uptime >> recon.txt
nica@nikasvs:~/isla-verde/docs$ cat recon.txt
Mon Aug 17 06:55:57 AM UTC 2026
nikasvs
 06:56:22 up 57 min,  1 user,  load average: 0.00, 0.01, 0.00
nica@nikasvs:~/isla-verde/docs$ _
```

This proves that I used manual pages, find, grep, diff, sudo, and chmod to investigate the system, verify changes, and manage file permissions.

EV-06: Finding, documentation, configuration change, and permissions

Proves manual use, diff, sudo accountability, chmod 600, and SSH status.

```
LS(1)                                     User Commands                               LS(1)
NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default). Sort entries alpha-
    betically if none of -cftuvSUX nor --sort is specified.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
        with -l, print the author of each file

    -b, --escape
        print C-style escapes for nongraphic characters

    --block-size=SIZE
        with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE
        format below

    -B, --ignore-backups
        do not list implied entries ending with ~

    -c
        with -lt: sort by, and show, ctime (time of last modification of file status infor-
        mation); with -l: show ctime and sort by name; otherwise: sort by ctime, newest
        first

Manual page ls(1) line 1 (press h for help or q to quit)
```

```
BSIT 4E Ubuntu Server 22.0 [Running] - Oracle VirtualBox
betically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.

-a, --all
    do not ignore entries starting with .

-A, --almost-all
    do not list implied . and ..

--author
    with -l, print the author of each file

-b, --escape
    print C-style escapes for nongraphic characters

--block-size=SIZE
    with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE
    format below

-B, --ignore-backups
    do not list implied entries ending with ~

-c      with -lt: sort by, and show, ctime (time of last modification of file status infor-
        mation); with -l: show ctime and sort by name; otherwise: sort by ctime, newest
        first

-C      list entries by columns

--color[=WHEN]
    colorize the output; WHEN can be 'always' (default if omitted), 'auto', or 'never';
    more info below

-d, --directory
    list directories themselves, not their contents

nica@nikasvs:~$ _
```

```
nica@nikasvs:~$ apropos "disk space"
df (1)          - report file system disk space usage
nica@nikasvs:~$ _
```

```
GNU nano 6.2 /etc/hosts *
127.0.0.1 localhost
127.0.1.1 nika_svs

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters

192.168.56.20
srv-isla01
```

```
nica@nikasvs:~$ diff /etc/hosts.bak /etc/hosts
9a10,12
>
> 192.168.56.20
> srv-isla01
```

```
nica@nikasvs:~/isla-verde$ ping -c 2 srv-isla01
PING srv-isla01 (192.168.56.20) 56(84) bytes of data.
ping: Warning: invalid tv_usec -88005735319406719 us
ping: Warning: time of day goes back (-37791039571350 s), taking countermeasures
ping: Warning: invalid tv_usec -88005735319404278 us
ping: Warning: time of day goes back (-37791039571350 s), taking countermeasures
64 bytes from srv-isla01 (192.168.56.20): icmp_seq=0 ttl=64 time=0.000 ms
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 14 8 0 57 ca 0 1 0 2 ae ba 82 6a 0 0 0 0 a5 3a b 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b
ping: Warning: invalid tv_usec -88005735319401004 us
ping: Warning: time of day goes back (-37791039571350 s), taking countermeasures
64 bytes from srv-isla01 (192.168.56.20): icmp_seq=0 ttl=64 time=0.000 ms (DUP!)
wrong data byte #16 should be 0x10 but was 0xc0
#16      c0 a8 38 14 8 0 57 ca 0 1 0 2 ae ba 82 6a 0 0 0 0 a5 3a b 0 0 0 0 0 10 11 12 13
#48      14 15 16 17 18 19 1a 1b

--- srv-isla01 ping statistics ---
2 packets transmitted, 1 received, +1 duplicates, 50% packet loss, time 1020ms
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 3
nica@nikasvs:~/isla-verde$ _
```

```
nica@nikasvs:~$ sudo -l
Matching Defaults entries for nica on nikasvs:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin,
    use_pty

User nica may run the following commands on nikasvs:
    (ALL : ALL) ALL
```

```

nica@nikasvs:~$ sudo tail -n 20 /var/log/auth.log
Aug 17 07:12:42 nikasvs sudo:      nica : TTY=tty1 ; PWD=/home/nica ; USER=root ; COMMAND=/usr/bin/ap
t install openssh-server
Aug 17 07:12:42 nikasvs sudo: pam_unix(sudo:session): session opened for user root(uid=0) by nica(ui
d=1000)
Aug 17 07:12:51 nikasvs useradd[2624]: new user: name=sshd, UID=113, GID=65534, home=/run/sshd, shel
l=/usr/sbin/nologin, from=none
Aug 17 07:12:51 nikasvs usermod[2631]: change user 'sshd' password
Aug 17 07:12:51 nikasvs chage[2638]: changed password expiry for sshd
Aug 17 07:12:52 nikasvs sshd[2766]: Server listening on 0.0.0.0 port 22.
Aug 17 07:12:52 nikasvs sshd[2766]: Server listening on :: port 22.
Aug 17 07:12:57 nikasvs sudo: pam_unix(sudo:session): session closed for user root
Aug 17 07:13:31 nikasvs sudo:      nica : TTY=tty1 ; PWD=/home/nica ; USER=root ; COMMAND=/usr/bin/fi
nd /etc -name *.conf
Aug 17 07:13:31 nikasvs sudo: pam_unix(sudo:session): session opened for user root(uid=0) by nica(ui
d=1000)
Aug 17 07:13:31 nikasvs sudo: pam_unix(sudo:session): session closed for user root
Aug 17 07:17:01 nikasvs CRON[2951]: pam_unix(cron:session): session opened for user root(uid=0) by (
uid=0)
Aug 17 07:17:01 nikasvs CRON[2951]: pam_unix(cron:session): session closed for user root
Aug 17 07:22:39 nikasvs sudo:      nica : TTY=tty1 ; PWD=/home/nica ; USER=root ; COMMAND=/usr/bin/cp
/etc/hosts /etc/hosts.bak
Aug 17 07:22:39 nikasvs sudo: pam_unix(sudo:session): session opened for user root(uid=0) by nica(ui
d=1000)
Aug 17 07:22:39 nikasvs sudo: pam_unix(sudo:session): session closed for user root
Aug 17 07:23:49 nikasvs sudo:      nica : TTY=tty1 ; PWD=/home/nica ; USER=root ; COMMAND=/usr/bin/na
no /etc/hosts
Aug 17 07:23:49 nikasvs sudo: pam_unix(sudo:session): session opened for user root(uid=0) by nica(ui
d=1000)
Aug 17 07:26:29 nikasvs sudo: pam_unix(sudo:session): session closed for user root
Aug 17 07:30:52 nikasvs sudo:      nica : TTY=tty1 ; PWD=/home/nica ; USER=root ; COMMAND=/usr/bin/ta
il -n 20 /var/log/auth.log
nica@nikasvs:~$

```

```

nica@nikasvs:~$ cd ~/isla-verde
nica@nikasvs:~/isla-verde$ ls -l backups/
total 8
-rw-r--r-- 1 nica nica 223 Aug 17 06:40 hosts.original
-rw-r--r-- 1 nica nica 386 Aug 17 06:40 os-release.original
nica@nikasvs:~/isla-verde$ _

```

```

nica@nikasvs:~/isla-verde$ chmod 600 backups/hosts.original
nica@nikasvs:~/isla-verde$ ls -l backups/hosts.original
-rw----- 1 nica nica 223 Aug 17 06:40 backups/hosts.original

```

```
nica@nikasvs:~/isla-verde$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-17 07:12:52 UTC; 21min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 2766 (sshd)
    Tasks: 1 (limit: 2187)
   Memory: 1.7M
      CPU: 13ms
   CGroup: /system.slice/ssh.service
           └─2766 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 17 07:12:52 nikasvs systemd[1]: Starting OpenBSD Secure Shell server...
Aug 17 07:12:52 nikasvs sshd[2766]: Server listening on 0.0.0.0 port 22.
Aug 17 07:12:52 nikasvs sshd[2766]: Server listening on :: port 22.
Aug 17 07:12:52 nikasvs systemd[1]: Started OpenBSD Secure Shell server.
nica@nikasvs:~/isla-verde$ _
```

RECONNAISSANCE REPORT

To: Supervisor, Isla Verde

From: Nica

Subject: Isla Verde Server Reconnaissance Report

Date: August 17, 2026

The Isla Verde server was inspected using standard Linux commands while logged in as the normal user *nica*. The hostname is *nikasvs*, verified using *hostname*, and the operating system is Ubuntu 22.04.5 LTS, verified using *cat /etc/os-release*.

The server has 2 processing units, shown by *nproc*, and an Intel Core i5-10400 @ 2.90 GHz, identified using *head -n 20 /proc/cpuinfo*. The system has approximately 1.9 GiB of RAM and 2.0 GiB of swap, according to *free -h. /proc/meminfo* reports the total memory as 1,986,200 kB. The *lsblk* command shows a 100G disk, with the main partition mounted at */*.

The */etc/passwd* file contains 33 accounts, confirmed using *wc -l /etc/passwd*. Most accounts use */usr/sbin/nologin*, as shown by the shell-counting pipeline. This indicates that many are system accounts rather than interactive users.

The */var/log/syslog* file contains 1,048 lines, according to *wc -l /var/log/syslog*. Recent system activity was viewed using *tail -n 20 /var/log/syslog*. The command *find /etc -name "*.conf" | wc -l* found 114 configuration files under */etc*.

The SSH service was checked using *systemctl status ssh* and was active and running on port 22. *grep -n PermitRootLogin /etc/ssh/sshd_config* found *#PermitRootLogin prohibit-password* on line 33. Because it is commented out, it is not an active setting by itself.

Before modifying */etc/hosts*, a backup was created using *sudo cp /etc/hosts /etc/hosts.bak*. The *diff* command confirmed that the *srv-isla01* entry was added. The backup *backups/hosts.original* was also protected with *chmod 600*, giving the owner read and write access while denying access to the group and others.

Recommendation

My recommendation is to continue using a backup-before-editing and verify-after-editing approach for system configuration files. Important files such as */etc/hosts* and *SSH configuration* files should always be backed up before changes are made. The system should also continue monitoring */var/log/auth.log* and other system logs because they provide useful evidence of administrative activity and possible problems. File permissions should be kept restrictive where appropriate instead of using unsafe permissions such as *777*.

REPORT QUESTIONS — ACTIVITIES 1–6

Activity 1:

ANSWER IN YOUR REPORT

1. What is the difference between what `uname -a` told you and what `/etc/os-release` told you?

ANSWER: `uname -a` shows information about the Linux kernel, it has kernel version, hostname, architecture, and build info. While `/etc/os-release` shows the operating system distribution such as the Ubuntu version and its codename.

2. Copy the exact error from WHOAMI. Which word did the system say was wrong?

ANSWER: WHOAMI: command not found. Because the Linux commands are case-sensitive.

Activity 2:

ANSWER IN YOUR REPORT

1. The Isla Verde web server will not start. Which directory do you look in first?

ANSWER: In the `/var/log` because this is where the system and application logs are stored.

2. You need to change the port SSH listens on. Which file holds that setting?

ANSWER: `/etc/ssh/sshd_config` is the SSH server configuration file.

3. Which file in `/var/log` changed most recently, and what command told you?

ANSWER: `auth.log` changed most recently in the screenshot. The command that showed this was `ls -lt | head`, which lists files by modification time with the newest first.

Activity 3:

ANSWER IN YOUR REPORT

1. `/etc/passwd` and `/var/log/syslog` have very different line counts. Which reading tool suits each, and why?

ANSWER: `/etc/passwd` is a short file, so `cat` is suitable because it can display the whole file at once.

`/var/log/syslog` is a much longer file with 1048 lines, so `less` is more suitable because it allows you to scroll through the file and search without displaying everything at once.

2. State your VM's total memory twice — once from `/proc/meminfo` and once from `free -h`. Why do the units differ but the numbers agree?

ANSWER:

- From `/proc/meminfo`: 1986200 kB
- From `free -h`: 1.9 GiB

It has the same total physical memory, but `/proc/meminfo` reports memory in kilobytes (kB), while `free -h` uses human-readable binary units (GiB) and rounds the value.

3. Why do most files in `/proc` show a size of zero even though reading them returns content?

ANSWER: `/proc` is a virtual filesystem. The files don't contain normal data stored on the disk. The contents are generated dynamically by the Linux kernel when you read them. Therefore, their reported file size can be 0 bytes even though they return information when read.

Activity 4:

ANSWER IN YOUR REPORT

1. The two cp commands printed nothing at all. Did they work? How did you confirm it, and which principle is this?

ANSWER: Yes, they worked. by running `ls -lh backups/`, that showed `hosts.original` and `os-release.original` in the backups directory. This demonstrates the “no news is good news” / silent success principle: many Linux commands produce no output when they complete successfully.

2. You are about to edit `/etc/ssh/sshd_config`. What is the exact command you run first, and why?

ANSWER: `sudo cp -i /etc/ssh/sshd_config ~/sshd_config.backup`

Because this can create a backup copy of the configuration file before editing it, so the original can be stored if it has a mistake.

Activity 5:

ANSWER IN YOUR REPORT

1. In plain words, what question does the five-stage pipeline answer about this server?

ANSWER: “Which login shells are used by the accounts on this server, how many accounts use each shell, and which shells are the most common?”

Results:

- `/usr/sbin/nologin` — 27 accounts
- `/bin/false` — 3 accounts
- `/bin/bash` — 2 accounts
- `/bin/sync` — 1 account

2. How many `.conf` files are under `/etc`? Is root login over SSH permitted — show the line and its number.

ANSWER: There are 114 `.conf` files under `/etc`.

The grep result shows:

- Line 33: `#PermitRootLogin prohibit-password`

Because the line starts with `#`, it is commented out, so it is not an active setting. The line itself does not enable root login. The effective SSH setting depends on the default/configuration applied elsewhere.

3. What is the difference between `>` and `>>`, and what does getting it wrong cost you?

ANSWER:

- `>` replaces/overwrites the existing file.
- `>>` appends the new output to the end of the file.

Getting `>` wrong when you meant `>>` can erase the existing contents of the file and replace them with the new output.

Activity 6:

ANSWER IN YOUR REPORT

1. Choose one command you have never seen — nl, column, watch, or basename. Using only man or --help, write one sentence saying what it does, name two options, and show it working. No internet.

ANSWER: I chose basename. It removes the directory part of a file path and displays only the file name.0

Two options are:

- -a — process multiple paths.
- -s SUFFIX — remove a specified suffix from the file name.

Example:

→ `basename /home/nica/isla-verde/docs/inventory-2026.txt`

Output:

→ `inventory-2026.txt`

2. What did diff show, and what would it have shown if you had accidentally deleted an existing line too?

ANSWER: diff showed that three new lines were added to /etc/hosts, including the 192.168.56.20 and srv-isla01 entry. The output 9a10,12 means lines 10–12 were added after line 9.

If I had accidentally deleted an existing line, diff would also show a deletion, using < to indicate the line that existed in the backup but was missing from the current file.

3. Find your own commands in the auth log. What does it mean for accountability that you cannot erase them?

ANSWER: The auth log records commands I ran with sudo, such as installing OpenSSH, running find, copying /etc/hosts, editing /etc/hosts, and reading the authentication log.

This provides accountability because the system records who used sudo, when they used it, and what command they ran. Even if a command changes the system, its activity can still be recorded in the authentication log.

4. Translate -rw----- into a sentence of ordinary English.

ANSWER: The owner can read and write the file, while the group and everyone else have no permissions.

REFLECTION

The command `cp /etc/hosts backups/hosts.original` behaved differently from what I expected because it did not display any message. At first, I thought the command might not have worked. I learned that many Linux commands remain silent when they complete successfully. I confirmed the copy by using `ls -lh backups/`, which showed the backup file.

The habit that will matter most to me is backing up before editing. The `/etc/hosts` activity showed me why this is important. Creating a backup allowed me to compare the original and modified files using `diff`. This can prevent mistakes from becoming bigger problems.

One thing I still do not fully understand is the `/proc` filesystem, especially why its files can show a size of zero even though they contain information when read. I understand that it is generated by the kernel, but I want to learn more about how Linux creates and updates this information.

Before you submit

✓	Check
✓	My logbook covers all six activities, not just the ones I found interesting.
✓	All six screenshots are present, numbered, captioned, and legible.
✓	Every figure in my report names the command that produced it.
✓	I backed up /etc/hosts before editing it, and my diff output is included.
✓	I did not work as root and ran nothing from the never-run list.
✓	My reflection names a real, specific confusion.

Grading Rubric · 100 Points

Find the description that best matches the work as a whole for each row, and award the points in that column. Where work falls between two columns, award the lower one and note why.

Criterion	Pts	Exemplary (100%)	Proficient (80%)	Developing (60%)	Beginning (40%)
1. Commands run correctly <i>LO1–LO4</i>	30	All activities completed with correct syntax, options, and paths. Outputs are as required. Any early errors were spotted and fixed without help.	Nearly all correct. One or two syntax or option errors, fixed after a prompt.	Several commands wrong, incomplete, or run in the wrong place. Some activities skipped. Errors left uncorrected.	Most commands missing, wrong, or evidently not run on the student's own machine.
2. Understands what the output means <i>LO1, LO2</i>	20	Navigates by both absolute and relative paths. States directory purposes correctly and interprets system output accurately, including the /proc readings.	Navigates correctly with minor hesitation. Interpretation accurate apart from one small error.	Navigation works but relies on one pattern only. Output is reported but not interpreted.	Cannot reliably reach required locations. Directory purposes wrong or copied word-for-word from the handout.
3. Search, pipes, and using the manuals <i>LO4, LO5</i>	20	Right tool chosen for each question. Pipeline works and each stage is explained in the student's own words. Unknown command solved from the machine alone, with the method shown.	Search tools used correctly for most questions. Pipeline works and is mostly explained. Unknown command solved, method described briefly.	Some questions answered, others guessed. Pipeline copied from the handout without explanation. Unknown command answered without showing method.	Search tools not used or used wrongly. No working pipeline. No evidence the manuals on the machine were used.
4. Safe practice <i>LO3, LO6</i>	15	Backed up before every change and can show it. Used sudo deliberately and can justify each use. Verified after every change. Worked as a normal user throughout.	Backed up before the main change. Used sudo appropriately. Verified most changes. Worked as a normal user.	Backed up only when told to. sudo used by habit without justification. Little verification after changes.	No backups. Worked as root or used chmod 777. Changes unverified, or ran something from the never-run list.
5. Logbook, evidence, and reflection <i>LO6</i>	15	Logbook covers all six activities with a genuine expectation recorded for each command. All six screenshots present and captioned. Reflection names a real, specific confusion. Another student could repeat the work from it.	Logbook covers all activities with minor gaps. Screenshots present and legible. Reflection answers all three prompts.	Logbook covers some activities only. Expectations left blank or filled in afterwards. Screenshots missing or unlabelled. Reflection generic.	Logbook fragmentary or absent. Screenshots mostly missing. Reflection missing or unrelated to the lab.

Criterion	Max	Score	Comments
1. Commands run correctly	30		
2. Understands what the output means	20		
3. Search, pipes, and using the manuals	20		
4. Safe practice	15		
5. Logbook, evidence, and reflection	15		
TOTAL	100		

Checked by: _____ Date: _____